

Week 16: Package Management (Part 2)

Date: June 22, 2026

Study Time: 5-8 hours total

GOAL

Manage packages on both Debian and Fedora-based systems

STUDY TASKS (1-2 hours)

Reading

☐ **"How Linux Works"** - Chapter 8 (RPM Package Management)

Video Learning

☐ **LearnLinuxTV** - "DNF Package Manager"

☐ **Professor Messer** - "RPM Package Management"

Concepts

☐ DNF vs YUM (history)

☐ RPM package format

☐ RPM vs DEB differences

☐ Repositories and .repo files

☐ EPEL (Extra Packages for Enterprise Linux)

☐ Package groups

☐ Transaction history

☐ Automatic updates

LAB WORK (3-4 hours)

Understanding DNF

All work in this section is on your **Fedora server**.

☐ **Check DNF version**

bash

```
ssh fedora
dnf --version
```

```
# DNF is the modern replacement for YUM
# YUM still works but redirects to DNF
```

☐ View configured repositories

bash

```
dnf repolist
# Shows enabled repos

dnf repolist all
# Shows all repos (enabled + disabled)

# Repository config files:
ls /etc/yum.repos.d/
cat /etc/yum.repos.d/fedora.repo
```

Basic DNF Commands

☐ Update package metadata

bash

```
sudo dnf check-update
# Checks for available updates
# Similar to 'apt update'
```

☐ Upgrade all packages

bash

```
sudo dnf upgrade
# OR
sudo dnf update # Same thing

# Upgrades all packages
# Similar to 'apt upgrade'
```

☐ Search for packages

bash

```
dnf search nginx

# More specific search
dnf search all nginx
# Searches in names, summaries, descriptions
```

☐ Show package information

bash

```
dnf info nginx

# Shows:
# - Name, Version, Release
# - Architecture
# - Size
# - Repository
# - Summary, Description
# - License
```

Installing Packages

☐ Install a package

bash

```
sudo dnf install tree

# Verify
tree --version
which tree
```

☐ Install multiple packages

bash

```
sudo dnf install htop ncdu curl wget
```

☐ Install without prompting (for scripts)

bash

```
sudo dnf install -y package_name
```

☐ Install specific version

```
bash

# List available versions
dnf list --showduplicates package_name

# Install specific version
sudo dnf install package_name-version
```

☐ Install from a .rpm file

```
bash

# Download a .rpm (example)
# wget https://example.com/package.rpm

# Install with dependencies
sudo dnf install ./package.rpm

# Or use rpm directly (doesn't resolve dependencies)
sudo rpm -ivh package.rpm
```

Removing Packages

☐ Remove a package

```
bash

sudo dnf remove package_name
# Removes package and dependencies not needed by others
```

☐ Remove unused dependencies

```
bash

sudo dnf autoremove
# Removes orphaned packages
```

☐ Clean cached data

```
bash
```

```
# Remove cached packages
```

```
sudo dnf clean packages
```

```
# Remove all cached data
```

```
sudo dnf clean all
```

Working with DNF

☐ List installed packages

```
bash
```

```
dnf list installed
```

```
# Search installed
```

```
dnf list installed | grep nginx
```

☐ List available packages

```
bash
```

```
dnf list available
```

```
# List from specific repo
```

```
dnf list available --repo=fedora
```

☐ Show package dependencies

```
bash
```

```
dnf repoquery --requires nginx
```

```
# What nginx needs
```

```
dnf repoquery --whatrequires nginx
```

```
# What needs nginx
```

☐ Which package provides a file?

```
bash
```

```
# Who provides /usr/bin/wget?  
dnf provides /usr/bin/wget  
  
# Who provides a command?  
dnf provides */ifconfig
```

DNF History

DNF tracks all transactions!

☐ View transaction history

```
bash
```

```
dnf history  
# Shows all installs, upgrades, removals
```

☐ View specific transaction

```
bash
```

```
dnf history info 5  
# Details of transaction #5
```

☐ Undo a transaction

```
bash
```

```
sudo dnf history undo 5  
# Undoes transaction #5  
# Be careful with this!
```

☐ Redo a transaction

```
bash
```

```
sudo dnf history redo 5
```

Package Groups

Groups are collections of related packages.

☐ List available groups

```
bash
```

```
dnf group list
```

```
# Or
```

```
dnf grouplist
```

☐ Show group contents

```
bash
```

```
dnf group info "Development Tools"
```

☐ Install a group

```
bash
```

```
sudo dnf group install "Development Tools"
```

```
# Installs compilers, build tools, etc.
```

☐ Remove a group

```
bash
```

```
sudo dnf group remove "Development Tools"
```

Working with RPM

RPM is the low-level package manager (like dpkg for Debian).

☐ Query all installed packages

```
bash
```

```
rpm -qa
```

```
# Lists all installed packages
```

```
rpm -qa | grep nginx
```

```
# Search installed
```

☐ Show package info

```
bash
```

```
rpm -qi nginx
```

```
# Detailed info about installed package
```

☐ List files in package

bash

```
rpm -ql nginx
# Lists all files installed by nginx
```

☐ Which package owns a file?

bash

```
rpm -qf /usr/sbin/nginx
# Shows: nginx-1.20.1-1.fc35.x86_64
```

☐ Install .rpm file directly

bash

```
sudo rpm -ivh package.rpm
# i = install, v = verbose, h = hash marks

# Upgrade package
sudo rpm -Uvh package.rpm

# Note: rpm doesn't resolve dependencies!
# Use dnf install ./package.rpm instead
```

☐ Remove with rpm

bash

```
sudo rpm -e package_name
# Erase (remove) package
```

Adding EPEL Repository

EPEL provides extra packages not in Fedora base.

☐ Install EPEL

```
bash
```

```
# On Fedora:  
sudo dnf install epel-release  
  
# Verify  
dnf repolist | grep epel
```

☐ Search EPEL packages

```
bash
```

```
dnf search --repo=epel htop
```

☐ Install from EPEL

```
bash
```

```
sudo dnf install htop  
# Now available from EPEL
```

Automatic Updates

☐ Install dnf-automatic

```
bash
```

```
sudo dnf install dnf-automatic
```

☐ Configure automatic updates

```
bash
```

```
sudo nano /etc/dnf/automatic.conf  
  
# Key settings:  
# apply_updates = yes    # Actually apply updates (not just download)  
# download_updates = yes  
# upgrade_type = security # Only security updates, or 'default' for all  
  
# Save and exit
```

☐ Enable the service

bash

```
# For daily automatic updates:
sudo systemctl enable --now dnf-automatic.timer

# Check status
systemctl status dnf-automatic.timer

# View timer schedule
systemctl list-timers dnf-automatic.timer
```

Compare: APT vs DNF

Create a comparison script:

☐ Side-by-side comparison

bash

```
nano ~/package-comparison.sh

#!/bin/bash

echo "=== Package Manager Comparison ==="
echo ""
echo "Task | Debian/Ubuntu (APT) | Fedora/RHEL (DNF)"
echo "-----|-----|-----"
echo "Update list | apt update | dnf check-update"
echo "Upgrade | apt upgrade | dnf upgrade"
echo "Install | apt install PKG | dnf install PKG"
echo "Remove | apt remove PKG | dnf remove PKG"
echo "Search | apt search PKG | dnf search PKG"
echo "Info | apt show PKG | dnf info PKG"
echo "List installed | dpkg -l | rpm -qa"
echo "Files in pkg | dpkg -L PKG | rpm -ql PKG"
echo "Which pkg owns file | dpkg -S FILE | rpm -qf FILE"
echo "Clean cache | apt clean | dnf clean all"
echo "Auto remove | apt autoremove | dnf autoremove"
echo "History | /var/log/apt/history.log | dnf history"

chmod +x ~/package-comparison.sh
./package-comparison.sh
```

Practical Exercise

☐ Install useful tools on Fedora

bash

```
ssh fedora

sudo dnf install -y \
    htop \
    ncdp \
    tree \
    curl \
    wget \
    vim \
    git \
    tmux \
    net-tools

# Verify installations
htop --version
tree --version
```

☐ Practice removal

bash

```
# Install test package
sudo dnf install cowsay
cowsay "Testing DNF!"

# Remove it
sudo dnf remove cowsay

# Verify removal
which cowsay # Should return nothing
```

Create Package Management Report

☐ DNF report script

bash

```
nano ~/dnf-report.sh
```

```
#!/bin/bash
```

```
echo "=== DNF Package Report ==="
```

```
echo "Generated: $(date)"
```

```
echo "Hostname: $(hostname)"
```

```
echo ""
```

```
echo "--- Installed Packages ---"
```

```
rpm -qa | wc -l
```

```
echo ""
```

```
echo "--- Available Updates ---"
```

```
dnf check-update | grep -v "^$" | wc -l
```

```
echo ""
```

```
echo "--- Enabled Repositories ---"
```

```
dnf repolist | grep -v "^$" | tail -n +2 | wc -l
```

```
echo ""
```

```
echo "--- Recent Transactions (last 5) ---"
```

```
dnf history | head -6
```

```
echo ""
```

```
echo "--- Largest Installed Packages ---"
```

```
rpm -qa --queryformat '%{SIZE} %{NAME}\n' | sort -rn | head -10 | \
    awk '{printf "%sMB %s\n", $1/1024/1024, $2}'
```

```
chmod +x ~/dnf-report.sh
```

```
./dnf-report.sh
```

FLASHCARD COMMANDS

1. `dnf`
2. `dnf update`
3. `dnf upgrade`
4. `dnf install`
5. `dnf remove`
6. `dnf search`
7. `dnf info`
8. `dnf list`
9. `dnf history`
10. `dnf autoremove`
11. `dnf clean all`
12. `rpm`
13. `rpm -qa`
14. `rpm -qi`
15. `rpm -ql`
16. `rpm -qf`
17. `rpm -ivh`

Concepts:

- DNF vs YUM?
 - RPM vs DEB?
 - What is EPEL?
 - Transaction history?
 - Package groups?
-

EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

NOTES

Packages installed on Fedora:

EPEL packages used:

Automatic updates configured?

☐ Yes

☐ No

Type: _____ (security only / all)

APT vs DNF comparison insights:

🎉 END OF MONTH 4 - Networking complete!

Completed: ☐

QUICK REFERENCE

DNF Command Summary:

bash

```
sudo dnf check-update      # Check for updates
sudo dnf upgrade           # Upgrade all packages
sudo dnf install PKG       # Install package
sudo dnf remove PKG        # Remove package
sudo dnf search PKG        # Search for package
sudo dnf info PKG          # Show package info
sudo dnf list installed    # List installed
sudo dnf autoremove        # Remove orphans
sudo dnf clean all         # Clear cache
dnf history                # View history
sudo dnf history undo ID   # Undo transaction
```

RPM Command Summary:

bash

```
rpm -qa                   # List all installed
rpm -qi PKG               # Package info
rpm -ql PKG               # List files
rpm -qf FILE              # Which package owns file
sudo rpm -ivh package.rpm # Install .rpm
sudo rpm -Uvh package.rpm # Upgrade .rpm
sudo rpm -e PKG           # Remove package
```

Package Format Differences:

Feature	DEB (Debian/Ubuntu)	RPM (Fedora/RHEL)
Extension	.deb	.rpm
High-level tool	apt	dnf
Low-level tool	dpkg	rpm
Repository format	sources.list	.repo files
Update database	apt update	dnf check-update
Transaction log	/var/log/apt/	dnf history

Config Locations:

- DNF: `/etc/dnf/dnf.conf`
- Repos: `/etc/yum.repos.d/`
- RPM database: `/var/lib/rpm/`
- Cache: `/var/cache/dnf/`

Previous Week: Package Management (apt)

Next Week: Security & Vulnerability Scanning